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UNIVERSITI
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UNIVERSITI TEKNOLOGI MARA
(UiTM) FACULTY OF COMPUTING,
INFORMATICS, AND MATHEMATICS

IMS566: ADVANCED WEB DESIGN DEVELOPMENT
AND CONTENT MANAGEMENT

ONLINE LEAVE MANAGEMENT SYSTEM

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Table of Contents

1.0	Introduction	1
2.0	GitHub Repository Link	1
3.0	Entity-Relationship Diagram (ERD)	2
3.1	Database Architecture Overview	2
3.1.1	Users Table	2
3.1.2	Leave Applications Table	2
4.0	System Requirements	3
4.1	Software & Version Compatibility	3
4.2	Dependencies & Package Configurations	3
5.0	User Interface Overview	4
6.0	Features & Functionalities	5
6.1	User Authentication	5
6.2	Main Dashboard Panel	5
6.3	Leave Request Portal & Listing Table	6
7.0	Workflow of Form (CRUD Cycle Illustration)	7
7.1	The Creation Phase (Create)	7
7.2	The Retrieval Phase (Read)	8
7.3	The Modification Phase (Update)	8
7.4	The Removal Phase (Delete)	9
8.0	Team Roles & Contributions	9
8.1	Database Architecture & Backend Engineering (Aidil)	9
8.2	Frontend Layout & Interface Design (Nazri)	10
8.3	Core CRUD Logic & Input Validation (Syahir)	10
8.4	Project Coordination & Quality Assurance (Amin)	10
9.0	Contact Information (Support)	10
10.0	Conclusion & Reflection	11

1.0 Introduction

The Online Leave Management System is a web-based administrative utility engineered to automate, simplify, and digitalize the traditional workflow of requesting and tracking employee time-off. By replacing inefficient paper forms and disjointed email threads with a centralized digital platform, the system eliminates communication bottlenecks and record-keeping errors while providing users with real-time transparency over their application status. Built on the Laravel MVC architecture with a lightweight local SQLite configuration, the platform establishes a secure full-stack environment that handles the complete CRUD lifecycle of leave requests. Through strict server-side validation and built-in authentication pipelines, the system maps user accounts to their respective applications via functional one-to-many relationships while enforcing strong ownership boundaries to protect data privacy. Ultimately, this project serves as a functional prototype demonstrating how modern web frameworks can be leveraged to transform backend database logic into a secure, practical, and highly usable organizational tool.

2.0 GitHub Repository Link

To ensure transparency, code quality verification, and seamless version control tracking, the complete source code for the Online Leave Management System has been hosted on GitHub. The repository includes all Laravel backend controllers, migration files, custom asset configurations, and frontend Blade layout templates.

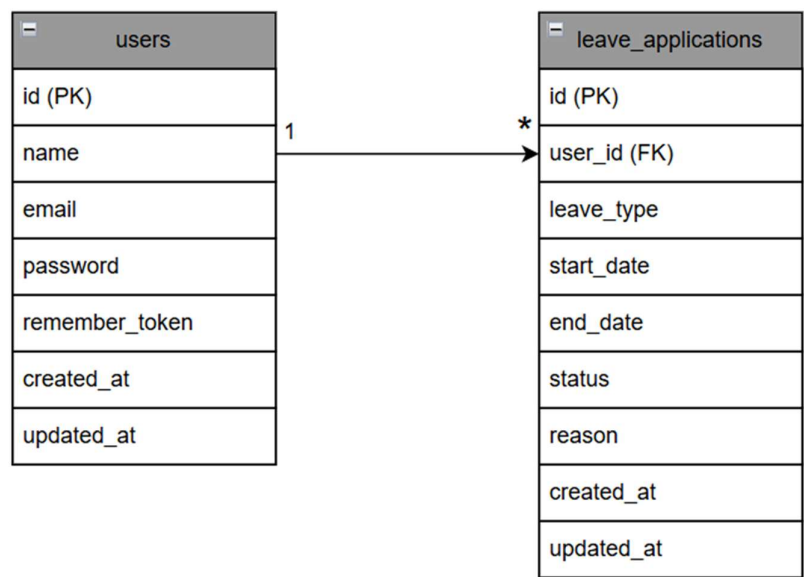
The project repository can be accessed directly via the following link:

- GitHub Repository URL: <https://github.com/amnaznn/leaveapp.git>

3.0 Entity-Relationship Diagram (ERD)

3.1 Database Architecture Overview

The database blueprint of the Online Leave Management System uses an Entity-Relationship Diagram (ERD) to map out the logical interaction between user credentials and transactional application records. By defining the exact attributes and structural links between tables, this diagram ensures complete data integrity and efficient information retrieval across the platform.



3.1.1 Users Table

The Users Table serves as the primary administrative directory for the system, storing essential authentication details and core user attributes required for secure account management. It encapsulates critical fields such as the auto-incrementing primary key, the user's name, a unique email address used as the login identifier, and an encrypted password string. This table acts as the foundational structural anchor, allowing the system to verify identities and manage active web sessions safely.

3.1.2 Leave Applications Table

The Leave Applications Table operates as a transactional ledger that tracks individual request elements linked to a specific user. It logs all relevant details for a time-off request, including the specific leave category, chronological start and end dates, a textual justification string, and the current approval status. To maintain structural integrity, this table includes a

foreign key column (user_id) that binds each application row back to its parent record in the Users Table, establishing a functional one-to-many relationship (1 →*) where an individual can submit multiple distinct requests while each request remains bound to exactly one account.

4.0 System Requirements

The Online Leave Management System relies on a modern development stack to ensure environmental stability, performance, and seamless execution across both local and production environments. The following breakdown outlines the core software parameters, version compatibilities, and dependencies required to deploy the system:

4.1 Software & Version Compatibility

- PHP (>= 8.2): The application leverages modern PHP features, including strict type declarations and performance enhancements, which are essential for running the latest versions of the Laravel framework.
- Laravel Framework (>= 10.x / 11.x): Serves as the primary full-stack architecture, utilizing the MVC design pattern, built-in ORM features, Eloquent, and authentication middleware.
- SQLite (>= 3.x): Chosen as the database engine for its zero-configuration, lightweight, and file-based nature, providing rapid testing and effortless local environment replication.
- Composer (>= 2.x): Utilized as the primary dependency manager to handle backend packages and PHP libraries smoothly.

4.2 Dependencies & Package Configurations

- Node.js (>= 18.x) & NPM / Vite: Used as the frontend compilation asset pipeline to manage, bundle, and hot-reload client-side assets and framework interfaces.
- Tailwind CSS: Integrated directly via the Vite asset compiler to construct a clean, utility-first user interface that renders responsively across both desktop and mobile viewports.
- Laravel Breeze (or Jetstream / Built-in Auth): Implemented to secure the platform by scaffolding complete, pre-configured authentication pipelines that handle secure user registration, session management, and password hashing automatically.

5.0 User Interface Overview

The Online Leave Management System features a modern, clean, and intuitive user interface designed to maximize usability and streamline administrative tasks. Built with a responsive layout, the interface effortlessly adapts to both desktop and mobile viewports, providing a consistent experience across all devices. Navigating the platform is straightforward, utilizing a persistent sidebar or top navigation menu that guides users smoothly through their workflow.

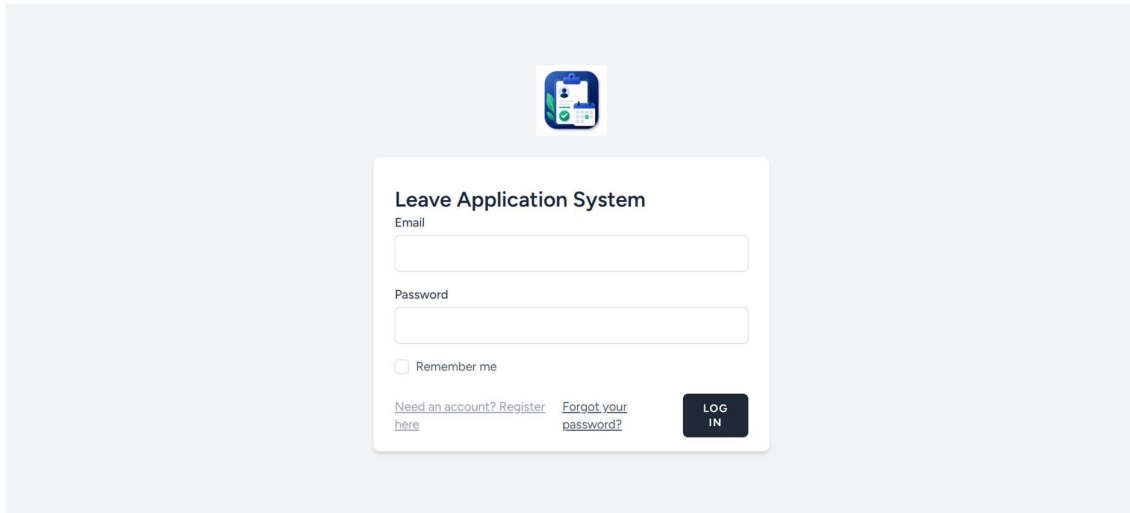
The user interface is structurally divided into key sections to organize essential tasks:

- **Authentication Portal:** A streamlined login and registration interface that provides secure entry gates, clear validation error feedback, and quick account creation paths.
- **User Dashboard:** The primary landing area post-login, presenting users with a clean overview of their profile summaries and a consolidated view of recent time-off updates.
- **Leave Application Form:** A dedicated, single-page form field designed for straightforward input entry, allowing users to select leave types, input custom dates, and provide justifications with minimal effort.
- **Request Management Ledger:** A structured tracking table displaying historical leave submissions, populated with distinct status badges (Pending, Approved, Rejected) that offer immediate visual updates on application progress.

6.0 Features & Functionalities

6.1 User Authentication

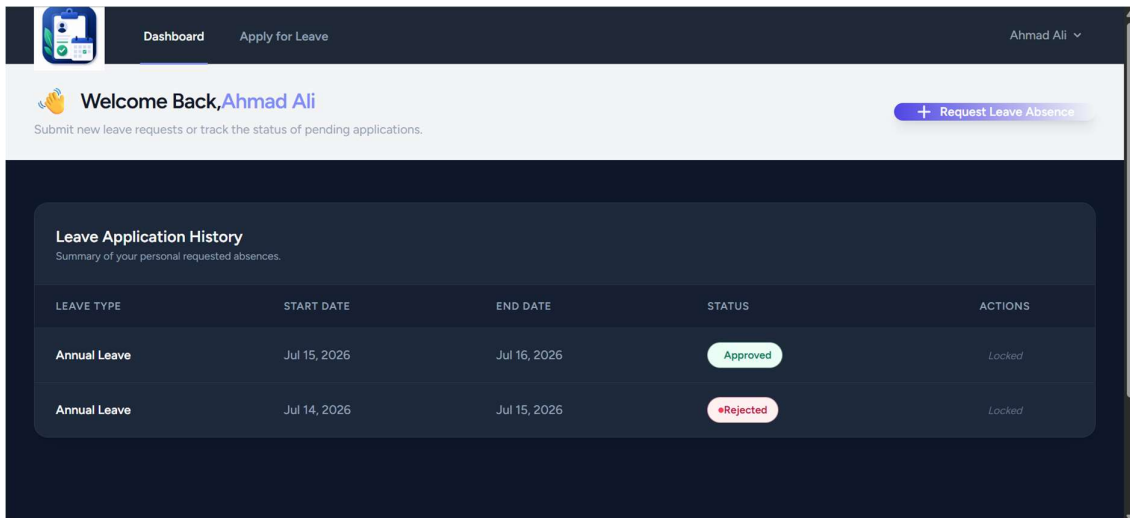
Secure registration and login screens powered by Laravel Breeze guard the application assets, ensuring users only access their personal data dashboards.



6.2 Main Dashboard Panel

The User Dashboard is a clean, modern interface designed to give employees an immediate snapshot of their time off status. At the top, a personalized greeting card welcomes the logged in user and provides a prominent, easy to use "+ Request Leave Absence" button for quick applications.

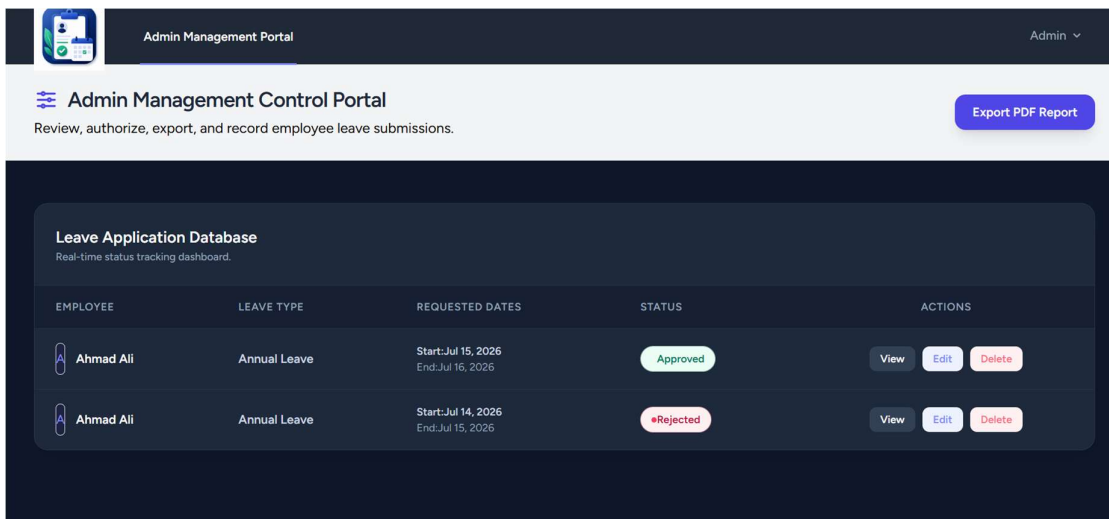
The main section features a structured Leave Application History table that tracks past requests. This table utilizes clear, color-coded status badges, such as green for Approved and red for Rejected, allowing users to monitor the outcomes of their submissions at a glance. Additionally, finalized requests show a "Locked" action state, preventing further modifications once management has decided.



6.3 Leave Request Portal & Listing Table

The Admin Management Control Portal is a centralized interface that allows HR administrators or managers to oversee all employee time off requests. At the top, a dedicated control banner features an "Export PDF Report" button, allowing administrators to instantly generate downloadable summary sheets for record keeping.

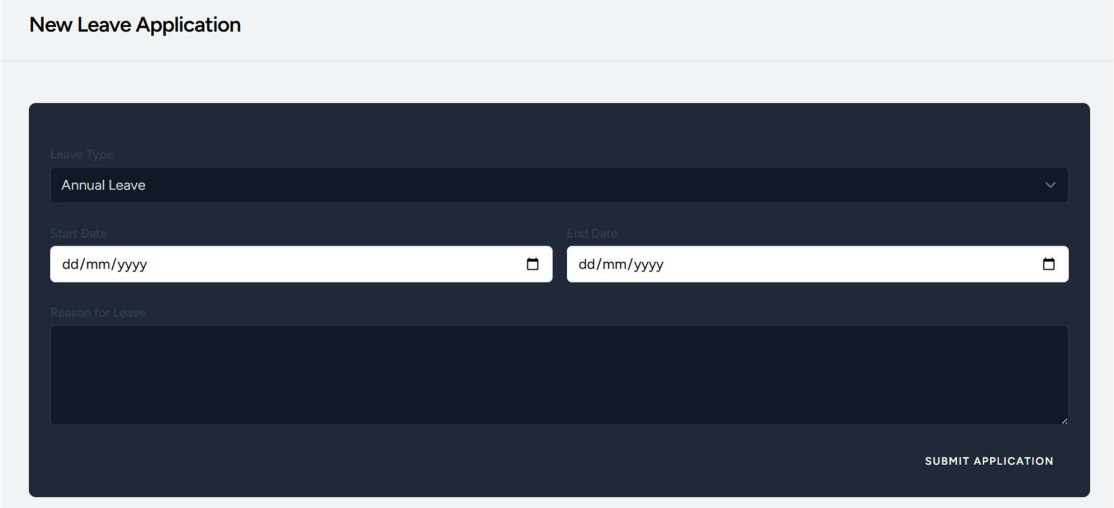
The main section contains a comprehensive Leave Application Database table that displays the employee names, leave types, and requested dates for the entire team. This dashboard uses clear color-coded status badges to show whether a request is approved or rejected, while giving administrators full control to view, edit, or delete any record in the system.



7.0 Workflow of Form (CRUD Cycle Illustration)

The operational core of the Online Leave Management System relies on the standardized CRUD (Create, Read, Update, Delete) data lifecycle matrix. This cyclical framework governs how information moves seamlessly between user interactions on the frontend interface and persistent data handling within the backend architecture. By utilizing Laravel's structured Model-View-Controller (MVC) pipelines, each phase of the workflow ensures strict data validation, secure routing protocols, and immediate feedback loops for the end user.

7.1 The Creation Phase (Create)



The screenshot displays a web form titled "New Leave Application". The form is set against a dark blue background with white text and input fields. It includes a "Leave Type" dropdown menu currently showing "Annual Leave". Below this are two date input fields labeled "Start Date" and "End Date", both with placeholder text "dd/mm/yyyy" and a calendar icon. A large text area for "Reason for Leave" is positioned below the date fields. At the bottom right of the form is a "SUBMIT APPLICATION" button.

The lifecycle initiates when an authenticated user submits the leave application form containing parameters like leave type, dates, and justification. The backend controller intercepts the payload, enforces server-side validation rules, binds the request to the active user's identifier, and commits it as a new record in the relational database.

7.2 The Retrieval Phase (Read)

Leave Application Detail

[← Back to Management Portal](#)

Ahmad Ali
Email: ahmadali@gmail.com

Approved

LEAVE TYPE	START DATE	END DATE
Annual Leave	2026-07-15	2026-07-16

REASON FOR LEAVE REQUEST

holiday

Edit Application

To provide data visibility, the engine queries the database using optimized lookups that selectively extract leave rows tied directly to the logged-in individual. This data is passed into the Blade layout compiler to dynamically generate a clean historical table complete with localized status badges (Pending, Approved, Rejected).

7.3 The Modification Phase (Update)

Edit Leave Application for Ahmad Ali

Leave Type

Annual Leave

Start Date

15/07/2026

End Date

16/07/2026

Management Status

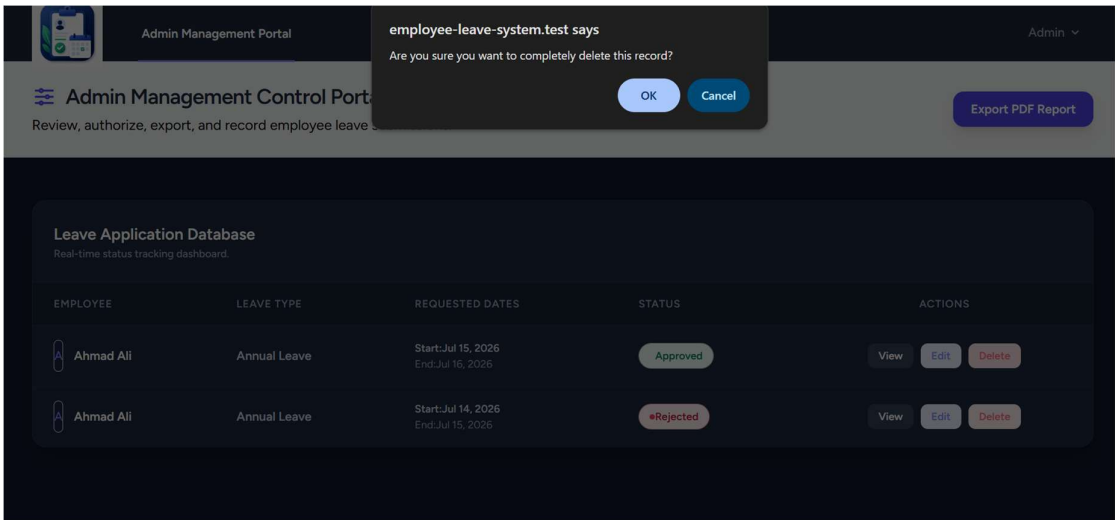
Approved

Save Changes

Cancel

If an entry requires scheduling adjustments, the admin selects the "Edit" trigger to pull the existing row values back into a pre-populated update form. Upon submitting changes, the modifications clear secondary validation checks before the controller updates the specific database columns and synchronizes the modification timestamps.

7.4 The Removal Phase (Delete)



If admin needs to retract a request, the cycle concludes with the deletion phase via a dedicated cancellation trigger. The controller intercepts the request, verifies that the admin owns the record, and purges the row from the database, instantly updating the dashboard view to maintain accurate system records

8.0 Team Roles & Contributions

The execution and successful deployment of the Online Leave Management System were achieved through a collaborative four-member project team structure consisting of Aidil, Nazri, Syahir, and Amin. By dividing engineering, design, and management requirements into distinct operational areas, the group ensured that database security, framework routing, and front-end interface adjustments were developed in parallel. This balanced delegation minimized integration bottlenecks and allowed for strict code quality controls across the entire development cycle.

8.1 Database Architecture & Backend Engineering (Aidil)

Aidil focused on the structural framework of the application, managing database design and the backend relational architecture. This role included defining the primary key constraints and setting up the migration scripts required to link users to their individual leave requests through functional one-to-many relationships. Additionally, Aidil implemented the underlying controller routing systems, ensuring that incoming server requests were mapped to their respective data endpoints while protecting system queries against common security vulnerabilities.

8.2 Frontend Layout & Interface Design (Nazri)

Nazri was responsible for the layout design and user interface polish, ensuring the application met modern web standards for clarity and navigation. Utilizing Tailwind CSS and Laravel Breeze components, Nazri styled the authentication pages, created responsive dashboard navigation paths, and configured clean visual accents, such as custom logo placements and high-visibility status badges. This work prioritized page scanning and layout responsiveness, ensuring administrative elements like data tables remained legible across varying device sizes.

8.3 Core CRUD Logic & Input Validation (Syahir)

Syahir directed the application's interactive workflow mechanics, focusing heavily on data validation pipelines and form security. This responsibility involved writing controller logic to intercept form actions, preventing users from inputting invalid date ranges, and establishing automated error message notifications. Syahir also engineered the ownership validation scripts, ensuring users could update or delete their personal records safely without exposing neighbouring table rows to unauthorized interference.

8.4 Project Coordination & Quality Assurance (Amin)

Amin acted as the project manager and lead quality tester, coordinating repository version control timelines and aligning system assets with academic criteria. This role focused on supervising feature integration, executing systematic interface testing to identify edge-case calculation bugs, and managing configuration properties during local environment testing. Amin also led the documentation compilation, translating the backend workflows and layout layouts into the final written report.

9.0 Contact Information (Support)

To ensure the long-term stability and reliability of the Online Leave Management System, a dedicated communications channel has been established to handle technical support and system maintenance inquiries. Because the platform operates as a centralized administrative utility, having a direct line of contact is essential for reporting runtime anomalies, clarifying deployment requirements, or addressing user account lockouts.

For all technical assistance, bug reporting, or system documentation inquiries, users and administrators can reach out directly via the primary project support email:

- Support & Maintenance Email: 2023695716@student.uitm.edu.my

All incoming communications are monitored by the development team to ensure that operational bugs are logged, security patches are deployed, and interface updates are rolled out systematically to preserve application integrity.

10.0 Conclusion & Reflection

The Online Leave Management System successfully demonstrates how the Laravel framework can be used to build a secure, practical administrative application. By utilizing a streamlined two-table database structure, the project proves that high utility and data integrity can be achieved without unnecessary architectural complexity. The platform successfully automates the leave request lifecycle while maintaining strict user ownership and input validation boundaries.

The development process provided the team with valuable, hands-on experience in full-stack engineering. Working with a local SQLite database configuration highlighted the efficiency of lightweight data models, while troubleshooting template assets and file paths deepened our understanding of real-world environment deployment.

While the current system fulfils all core assignment requirements, its modular architecture is fully prepared for future expansion. The platform can easily scale to include multi-authentication roles, automated email notifications, and multi-tier approval workflows, making it a reliable foundation for broader organizational needs.